

# Curriculum Vitae

## Nachiket Ganesh Kapre

Date of Birth: September 2nd, 1980

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### Employment Record

**Visiting Fellow**, Tenstorrent LLC., Jan 2026-present

**Professor**, Department of Electrical and Computer Engineering, University of Waterloo, Canada, July 2026 (official start date)

**Associate Professor**, Department of Electrical and Computer Engineering, University of Waterloo, Canada, July 2020-July 2026

**Research Director**, Xilinx/AMD, Singapore, Jan 2022-Feb 2023

**Assistant Professor**, Department of Electrical and Computer Engineering, University of Waterloo, Canada, Oct 2016-June 2020

**Chief Technology Officer**, Plunify Inc, 2014-2016

**Assistant Professor**, Nanyang Technological University, Singapore, Oct 2012–Sept 2016

**Consultant** Maxeler Inc. (bought by Groq, and then by NVIDIA), 2011–2012

**Junior Research Fellow** Imperial College London, Oct 2010–Sept 2012

**Visiting Graduate Student** University of Pennsylvania, Fall 2006–Fall 2010

**Summer Intern** Xilinx Inc., Summer 2005

**Research Assistant** ICLab Lab (Caltech), Fall 2004–Spring 2006

**Research Assistant** Koch Lab (Caltech), Spring 2004

**Design Engineer** Paxonet Communications Inc. (acquired by Conexant, followed by Synaptics), 2002–2003

### Leave of Absence

**Parental Leave**, Dec 2019-Apr 2019

**Leave of Absence**, Jan 2022-Feb 2023

## Education

**Ph.D. Computer Science**, California Institute of Technology,  
September 2010

Thesis: *SPICE<sup>2</sup>: A Spatial Parallel Architecture for Accelerating the SPICE Circuit Simulator*

**M.S. Computer Science**, California Institute of Technology  
June 2006

Thesis: *Packet-Switched FPGA-Overlay Networks*

**M.S. Electrical Engineering**, California Institute of Technology  
June 2005

**B.E. Electronics and Telecommunication Engineering**, University of Pune, India  
August 2002

Thesis: *FPGA-based Testing System for Siemens Railway Signalling Relays*

Ranked 1st in a class of 773 students in the Sophomore, Junior, and Senior years

## Publications

### Preprints

#### **“A Protocol-Independent Transport Architecture”**

Kimiya Mohammadtaheri, David Gao, Samuel Zhang, Matthew Chen, Eric Su, Pengyu Ji, Saad Syed, Chris Neely, Mario Baldi, Nachiket Kapre, and Mina Tahmasbi Arashloo  
*arXiv:2605.02210*, May 2026

### Peer-Reviewed Journal Articles

The top refereed journal in the FPGA/Reconfigurable computing space is ACM TRETs. Academics in this area typically publish papers at conferences and sometimes in journals. Extended versions of the top few papers from the prestigious FPGA conferences are generally submitted to ACM TRETs. I have five journal papers in TRETs, two of which received Best Paper Awards for 2017 and 2023. Graduate and undergraduate student authors supervised by me are identified with an underline

#### J12. **“HopliteML: Evolving application customized FPGA NoCs with adaptable routers and regulators”**

Gurshaant Malik, Ian Elmor Lang, Rodolfo Pellizzoni, Nachiket Kapre  
*ACM Transactions on Reconfigurable Technology and Applications*  
**(FPL 2020 Stamatis Vassiliadis Best Paper Nominee, invited to submit extended journal article, peer-reviewed.)**

#### J11. **“RapidLayout: Fast Hard Block Placement of FPGA-optimized Systolic Arrays using Evolutionary Algorithms”**

Niansong Zhang, Xiang Chen, Nachiket Kapre  
*ACM Transactions on Reconfigurable Technology and Applications*  
**(FPL 2020 Michal Servit Best Paper Nominee, invited to submit extended journal article, peer-reviewed.)**  
**(TRETs 2023 Best Paper Award)**

#### J10. **“HopliteBuf: Network Calculus-based design of FPGA NoCs with Provably Stall-Free FIFOs”**

Tushar Garg, Saud Al Wasly, Rodolfo Pellizzoni, Nachiket Kapre  
*ACM Transactions on Reconfigurable Technology and Applications*  
**(FPGA 2019 Best Paper Nominee, invited to submit extended journal article, peer-reviewed.)**

#### J9. **“CaffePresso: Accelerating Convolutional Networks on Embedded SoCs”**

Gopalakrishna Hegde Siddhartha, Nachiket Kapre  
*ACM Transactions on Embedded Computing Systems Volume 17 Issue 1 Page 15:1-15:26* January 2018

#### J8. **“Hoplite: A Deflection-Routed Directional Torus NoC for FPGAs”**

Nachiket Kapre, Jan Gray  
*ACM Transactions on Reconfigurable Technology and Systems Volume 10 Issue 2 Page 14:1-14:24*, April 2017 **(TRETs 2017 Best Paper Award)**

#### J7. **“Optimizing Soft Vector Processing in FPGA-based Embedded Systems”**

Nachiket Kapre  
*ACM Transactions on Reconfigurable Technology and Systems Volume 9 Issue 3 Page 17:1-17:17*, September 2016

- J6. **"A Case for Embedded FPGA-based SoCs in Energy-Efficient Acceleration of Graph Problem"**  
Pradeep Moorthy, Nachiket Kapre  
*Supercomputing Frontiers and Innovations (Special Best Papers Issue) Volume 2 Number 3 Page 76-86, November 2015*
- J5. **"Communication Optimization of Iterative Sparse Matrix-Vector Multiply on GPUs and FPGAs"** (11 pages)  
Abid Rafique, George Constantinides, Nachiket Kapre  
*IEEE Transactions on Parallel and Distributed Systems Volume 26 Issue 1 Page 24-34, January 2015*
- J4. **"SPICE<sup>2</sup> - Spatial Processors Interconnected for Concurrent Execution for acceleration the SPICE Circuit Simulator using an FPGA"** (14 pages)  
Nachiket Kapre, and André DeHon  
*in IEEE Transactions in Computer Aided Design of Integrated Circuits and Systems (Special Issue on Parallel CAD) Volume 31 Issue 1 Page 9-22 January 2012*
- J3. **"An NoC Traffic Compiler for efficient FPGA implementation of Sparse Graph-Oriented Workloads"** (12 pages)  
Nachiket Kapre, and André DeHon  
*International Journal of Reconfigurable Computing Volume 2011 Article ID 745147 (Open-Access article)*
- J2. **"Spatial Hardware Implementation for Sparse Graph Algorithms in GraphStep"** (20 pages)  
Michael deLorimier, Nachiket Kapre, Nikil Mehta and André DeHon  
*ACM Transactions on Autonomous and Adaptive Systems: Spatial Computing Special Issue Volume 6 Issue 3 Page 17:1-17:20 September 2011*
- J1. **"Pipelined Saturated Accumulation"** (11 pages)  
Karl Papadantonakis, Nachiket Kapre, Stephanie Chan, and André DeHon  
*IEEE Transactions on Computers Volume 58 Issue 2 Page 208-219 February 2009*

### *Peer-Reviewed Conference Articles (Full)*

Conferences are the primary venue for publishing and scholarship in the Reconfigurable computing domain. There are four top refereed conferences in the field in a calendar year: ACM/SIGDA International Symposium on Field-Programmable Gate Arrays (FPGA), IEEE International Symposium on Field-Programmable Custom Computing Machines (FCCM), International Conference on Field-Programmable Logic and Applications (FPL) , and IEEE International Conference on Field Programmable Technology (FPT) . I have extensively published in all these conferences over the years and won Best Paper Awards at FPT 2011 and FPL 2015. I have highlighted most significant papers with [MSP](#) tag. Graduate and undergraduate student authors supervised by me are identified with an underline.

- C56. **"Jack of All Scales: A Versatile FPGA Tensor Block for MXFP Precisions"** (9 pages)  
Marwan Mekhemer, Ahmed Elsousy, Balaji Venkatesh, Raphael Rowley, Vaughn Betz, Nachiket Kapre, and Andrew Boutros  
*International Conference on Field-Programmable Logic and Applications, Sep 2026*
- C55. **"Cocotb-PYNQ-PR: From Co-Simulation to Deployment, A Unified DFX Framework for PYNQ"** (9 pages)  
Emir Guevara, Chaitanya Sharma, Gavin Lusby, and Nachiket Kapre  
*International Conference on Field-Programmable Logic and Applications, Sep 2026*  
**(Stamatis Vassiliadis Memorial Best Paper Award)**

- C54. **"GraphNoC: Graph Neural Networks for Application-Specific FPGA NoC Performance Prediction"**<sup>MSP</sup> (8 pages)  
Gurshaant Malik, Nachiket Kapre  
*International Symposium on Field Programmable Technology*, Dec 2024  
**(Best Paper Award)**
- C53. **"Ditty: Directory-based cache coherence for multicore safety-critical systems"** (6 pages)  
Zhuanhao Wu, Marat Bekmyrza, Nachiket Kapre, Hiren Patel  
*Design, Automation & Test in Europe Conference & Exhibition*, Apr 2023  
**(Best Paper Nominee)**
- C52. **"Managing HBM Bandwidth on Multi-Die FPGAs with FPGA Overlay NoCs"** (8 pages)  
Srinirdheeshwar Kuttuva Prakash, Hiren Patel, Nachiket Kapre  
*International Conference on Field Programmable Custom Computing Machines*, May 2022
- C51. **"Mocarabe: High-Performance Time-Multiplexed Overlays for FPGAs"** (8 pages)  
Frederick Tombs, Alireza Mellat, Nachiket Kapre  
*International Conference on Field Programmable Custom Computing Machines*, May 2021
- C50. **"RapidLayout: Fast Hard Block Placement of FPGA-optimized Systolic Arrays using Evolutionary Algorithms"**<sup>MSP</sup> (8 pages)  
Niansong Zhang, Xiang Chen, Nachiket Kapre  
*International Symposium on Field Programmable Logic and Applications*, Sep 2020  
**(Michal Servit Best Paper Nominee)**
- C49. **"Learn the Switches: Evolving FPGA NoCs with Stall-Free and Backpressure based Routers"** (8 pages)  
Gurshaant Malik, Ian Lang, Rodolfo Pellizoni, Nachiket Kapre  
*International Symposium on Field Programmable Logic and Applications*, Sep 2020  
**(Stamatis Vassiliadis Best Paper Nominee)**
- C48. **"Partitioning Systolic Arrays for Fun and Profit"** (8 pages)  
Harry Chan, Gurshaant Malik, Nachiket Kapre  
*International Symposium on Field Programmable Technology*, Dec 2019  
**(Best Paper Award)**
- C47. **"Timing-aware routing in the RapidWright framework"** (7 pages)  
Leo Liu, Nachiket Kapre  
*International Symposium on Field Programmable Logic and Applications*, Sep 2019
- C46. **"Scaling the Cascades: Interconnect-aware FPGA implementation of Machine Learning problems"** (8 pages)  
Ananda Samajdar, Tushar Garg, Tushar Krishna, Nachiket Kapre  
*International Symposium on Field Programmable Logic and Applications*, Sep 2019
- C45. **"Enhancing Butterfly Fat Tree NoCs for FPGAs with lightweight flow control"** (8 pages)  
Gurshaant Malik, Nachiket Kapre  
*International Conference on Field Programmable Custom Computing Machines*, April 2019
- C44. **"HopliteBuf: FPGA NoCs with Provably Stall-Free FIFOs"**<sup>MSP</sup> (10 pages)  
Tushar Garg, Saud Wasly, Rodolfo Pellizoni, Nachiket Kapre  
*International Symposium on Field-Programmable Gate Arrays*, Feb 2019  
**(Best Paper Nominee)**, Invited to FPGA 2019 Special Best Papers Issues at ACM TRETs

- C43. **"Implementing NEF Neural Networks on Embedded FPGAs"** (8 pages)  
Benjamin Morcos, Terrence Stewart, Chris Eliasmith, Nachiket Kapre  
*International Conference on Field Programmable Technology*, December 2018
- C42. **"DaCO: A High-Performance Token Dataflow Coprocessor Overlay for FPGAs"** (8 pages)  
Siddhartha, Nachiket Kapre  
*International Conference on Field Programmable Technology*, December 2018
- C41. **"FastTrack: Leveraging Heterogeneous FPGA Wires to Design Low-Cost High-Performance Soft NoCs"**<sup>MSP</sup> (13 pages)  
Nachiket Kapre, Tushar Krishna  
*International Symposium on Computer Architecture*, June 2018
- C40. **"LegUp-NoC: High-Level Synthesis of Loops with Indirect Addressing"** (8 pages)  
Asif Islam, Nachiket Kapre  
*International Conference on Field Programmable Custom Computing Machines*, April-May 2018
- C39. **"Hoplite-Q: Priority-Aware Routing in FPGA Overlay NoCs"** (8 pages)  
Siddhartha, Nachiket Kapre  
*International Conference on Field Programmable Custom Computing Machines*, April-May 2018
- C38. **"HopliteRT: An Efficient FPGA NoC for Real-Time Applications"** (8 pages)  
Saud Al Wasly, Rodolfo Pellizzoni, Nachiket Kapre  
*International Conference on Field Programmable Technology*, December 2017
- C37. **"Deflection Routed Butterfly Fat Trees on FPGAs"** (8 pages)  
Nachiket Kapre  
*International Symposium on Field Programmable Logic and Applications*, Sep 2017
- C36. **"Enabling Partial Reconfiguration and Low Latency Routing using Segmented FPGA NoCs"** (8 pages)  
Kizhepatt Vipin, Jan Gray, Nachiket Kapre  
*International Symposium on Field Programmable Logic and Applications*, Sep 2017
- C35. **"On Bit-Serial NoCs for FPGAs "** (8 pages)  
Nachiket Kapre  
*International Symposium on Field Programmable Custom Computing Machines*, May 2017
- C34. **"Implementing FPGA overlay NoCs using the Xilinx UltraScale memory cascades"** (8 pages)  
Nachiket Kapre  
*International Symposium on Field Programmable Custom Computing Machines*, May 2017
- C33. **"eBSP: Managing NoC traffic for BSP workloads on the 16-core Adapteva Epiphany-III Processor"** (6 pages)  
Siddhartha, Nachiket Kapre  
*Design, Automation, and Test in Europe*, March 2017
- C32. **"Deflection Routing for Multi-Level FPGA Overlay NoCs"** (8 pages)  
Chethan Kumar H B, Shubham Agarwal, Nachiket Kapre  
*International Conference on Field Programmable Technology*, December 2016
- C31. **"Preventive Detection of Mosquito Populations using Embedded Machine Learning on Low Power IoT Platforms"** (10 pages)  
Prashant Ravi, Uma Syam, Nachiket Kapre  
*Seventh ACM Symposium on Computing and Development*, Nov 2016

- C30. **"CaffePresso: An Optimized Library for Deep Learning on Embedded Accelerator-based platforms"**<sup>MSP</sup> (10 pages)  
Gopalakrishna Hegde, Siddhartha, Nachiappan Ramasamy, Nachiket Kapre  
*International Conference on Compilers, Architecture, and Synthesis for Embedded Systems*, Oct 2016  
**(Best Paper Award)**
- C29. **"Hoplite-DSP: Harnessing the Xilinx DSP48 Multiplexers to efficiently support NoCs on FPGAs"** (10 pages)  
Chethan Kumar H B, Nachiket Kapre  
*International Conference on Field Programmable Logic and Applications*, Sep 2016
- C28. **"Boosting Convergence of Timing Closure using Feature Selection in a Learning-Driven Approach"** (9 pages)  
Que Yanghua, Harnhua Ng, Nachiket Kapre  
*International Conference on Field Programmable Logic and Applications*, Sep 2016
- C27. **"Survey of Domain-Specific Languages for FPGA Computing"** (10 pages)  
Nachiket Kapre, Samuel Bayliss  
*International Conference on Field Programmable Logic and Applications*, Sep 2016
- C26. **"Marathon: Statically-Scheduled Conflict-Free Routing on FPGA Overlay NoCs"** (8 pages)  
Nachiket Kapre  
*International Symposium on Field Programmable Custom Computing Machines*, May 2016
- C25. **"GPU-Accelerated High-Level Synthesis for Bitwidth Optimization of FPGA Datapaths"** (8 pages)  
Nachiket Kapre, Ye Deheng  
*International Symposium on Field-Programmable Gate Arrays*, Feb 2016
- C24. **"Hoplite: Building Austere Overlay NoCs for FPGAs"**<sup>MSP</sup> (6 pages)  
Nachiket Kapre, Jan Gray  
*International Conference on Field Programmable Logic and Applications*, Sep 2015  
**(Michael Servit Best Paper Award)**
- C23. **"Limits of FPGA Acceleration of 3D Green's Function Computation for Geophysical Applications"** (6 pages)  
Nachiket Kapre, Selvakumar Jayakrishnan, Parjanya Gupta, Sagar Masuti, Sylvain Barbot  
*International Conference on Field Programmable Logic and Applications*, Sep 2015
- C22. **"Custom FPGA-based Soft-Processors for Sparse Graph Acceleration"** (8 pages)  
Nachiket Kapre  
*International Conference on Application-Specific Systems, Architectures and Processors*, July 2015
- C21. **"Zedwulf: Power-Performance Tradeoffs of a 32-node Zynq SoC cluster"** (8 pages)  
Pradeep Moorthy, Nachiket Kapre  
*International Symposium on Field Programmable Custom Computing Machines*, May 2015
- C20. **"Driving Timing Convergence of FPGA Designs through Machine Learning and Cloud Computing"**<sup>MSP</sup> (8 pages)  
Nachiket Kapre, Bibin Chandrashekharan, Harnhua Ng, Kirvy Teo  
*International Symposium on Field Programmable Custom Computing Machines*, May 2015
- C19. **"Energy-Efficient Acceleration of OpenCV Saliency Computation using Soft Vector Processors"** (8 pages)

Gopalakrishna Hegde, Nachiket Kapre

*International Symposium on Field Programmable Custom Computing Machines*, May 2015

- C18. **"On Data Forwarding in Deeply Pipelined Soft Processor"** (8 pages)  
Hui Yan Cheah, Suhaib A. Fahmy and Nachiket Kapre  
*International Symposium on Field-Programmable Gate Arrays*, Page 181-189, February 2015
- C17. **"Relax-Miracle: GPU Parallelization of Semi-Analytic Fourier-Domain solvers for Earthquake Modeling"** (8 pages)  
 Sagar Masuti, Sylvain Barbot, and Nachiket Kapre  
*International Conference on High Performance Computing* December 2014
- C16. **"Comparing Soft and Hard Vector Processing in FPGA-based Embedded Systems"** (6 pages)  
 Soh Jun Jie, and Nachiket Kapre  
*International Conference on Field Programmable Logic and Applications*, Page 1-7, September 2014  
**(Best Paper Nominee)**
- C15. **"System-Level FPGA Device Driver with High-Level Synthesis Support"** (8 pages)  
 Vipin Kizhepatt, Shreejit Shanker, Dulitha Gunasekara, Suhaib A Fahmy, Nachiket Kapre  
*International Conference on Field Programmable Technology*, Page 128-135, December 2013
- C14. **"Exploiting Input Parameter Uncertainty for Reducing Datapath Precision of SPICE Device Models"** (8 pages)  
 Nachiket Kapre  
*IEEE Symposium on Field-Programmable Custom Computing Machines*, Page 189-197, April 2013
- C13. **"Application Composition and Communication Optimization of Iterative Solvers using FPGAs"** (8 pages)  
 Abid Rafique, Nachiket Kapre and George Constantinides  
*IEEE Symposium on Field-Programmable Custom Computing Machines*, Page 153-160, April 2013  
**(HiPEAC Paper Award)**
- C12. **"Enhancing Performance of Tall-Skinny QR factorization"** (8 pages)  
 Abid Rafique, Nachiket Kapre, and George Constantinides  
*International Conference on Field Programmable Logic and Applications*, Page 443-450, August 2012
- C11. **"A High Throughput FPGA-based Implementation of the Lanczos Method for the Symmetric Extremal Eigenvalue Problem"** (12 pages)  
 Abid Rafique, Nachiket Kapre, and George Constantinides  
*International Symposium on Applied Reconfigurable Computing*, Page 239-250, March 2012
- C10. **"FX-SCORE: A Framework for Fixed-Point Compilation of SPICE Device Models using Gappa++"** (8 pages)  
 Helene Martorell, and Nachiket Kapre  
*IEEE Symposium on Field-Programmable Custom Computing Machines*, Page 77-84, April 2012
- C9. **"VLIW-SCORE: Beyond C for Sequential Control of SPICE FPGA Acceleration"**<sup>MSP</sup> (9 pages)  
 Nachiket Kapre, and André DeHon  
*International Conference on Field Programmable Technology*, Page 1-9, December 2011  
**(Best Paper Award)**

- C8. **"Parallelizing Sparse Matrix-Solve for SPICE Circuit Simulation using FPGAs"** (9 pages)  
Nachiket Kapre, and André DeHon  
*International Conference on Field Programmable Technology*, Page 190-198, December 2009
- C7. **"Performance Comparison of Single-Precision SPICE Model-Evaluation on FPGA, GPU, Cell, and Multi-Core Processors"** (8 pages)  
Nachiket Kapre, and André DeHon  
*International Conference on Field Programmable Logic and Applications*, Page 65-72, September 2009
- C6. **"Acceleration SPICE Model-Evaluation using FPGAs"** (8 pages)  
Nachiket Kapre, and André DeHon  
*IEEE Symposium on Field-Programmable Custom Computing Machines*, Page 37-44, April 2009
- C5. **"Optimistic Parallelization of Floating-Point Accumulation"** (9 pages)  
Nachiket Kapre, and André DeHon  
*IEEE Symposium on Computer Arithmetic*, Page 205-216, June 2007
- C4. **"Packet-Switched vs. Time-Multiplexed FPGA Overlay Networks"** (10 pages)  
Nachiket Kapre, Nikil Mehta, Michael deLorimier, Raphael Rubin, Henry Barnor, Michael Wilson, Michael Wrighton, and André DeHon  
*IEEE Symposium on Field-Programmable Custom Computing Machines*, Page 205-216, April 2006  
**(FCCM20 25-most Influential Papers Award)**
- C3. **"GraphStep: A System Architecture for Sparse Graph Algorithms"** (9 pages)  
Michael deLorimier, Nachiket Kapre, Nikil Mehta, Dominic Rizzo, Ian Eslick, Raphael Rubin, Tomas Uribe, Thomas Knight Jr., and André DeHon  
*IEEE Symposium on Field-Programmable Custom Computing Machines*, Page 143-151, April 2006
- C2. **"Pipelined Saturated Accumulation"** (11 pages)  
Karl Papadantonakis, Nachiket Kapre, Stephanie Chan, and André DeHon  
*International Conference on Field-Programmable Technology*, Page 19-26, December 2005
- C1. **"Design Patterns for Reconfigurable Computing"** (11 pages)  
André DeHon, Joshua Adams, Michael deLorimier, Nachiket Kapre, Yuki Matsuda, Helia Naeimi, Michael Vanier, and Michael Wrighton  
*IEEE Symposium on Field-Programmable Custom Computing Machines*, Page 13-23, April 2004

#### *Peer Reviewed Conference Articles (Short)*

- S16. **"Cocotb-Pynq: Co-simulating Python+RTL applications targeting Pynq platforms with Cocotb"** (4 pages)  
Gavin Lusby, Nachiket Kapre  
*International Conference on Field Programmable Logic and Applications*, Sep 2025
- S15. **"Worst-Case Latency Analysis for the Versal NoC Network Packet Switch"** (6 pages)  
Ian Lang, Nachiket Kapre, Rodolfo Pellizzoni  
*IEEE/ACM Symposium on Networks-on-Chip*, Oct 2021
- S14. **"Exploring the Impact of Switch Arity on Butterfly Fat Tree FPGA NoCs"** (4 pages)  
Ian Lang, Ziqiang Huang, Nachiket Kapre  
*International Conference on Field Programmable Custom Computing Machines*, April 2020

- S13. **"RapidRoute: Fast Assembly of Communication Structures for FPGA Overlays"** (4 pages)  
 Leo Liu, Jay Weng, Nachiket Kapre  
*International Conference on Field Programmable Custom Computing Machines*, April 2019  
**(Best Short Paper Runner-Up)**
- S12. **"Applying Models of Computation to OpenCL Pipes for FPGA Computing"** (4 pages)  
 Nachiket Kapre, Hiren Patel  
*5th International Workshop on OpenCL*, May 2017
- S11. **"120-core microAptiv MIPS Overlay for the Terasic DE5-NET FPGA board"** (6 pages)  
 Chethan Kumar H B, Gourav Modi, Prashant Ravi, Nachiket Kapre  
*International Symposium on Field Programmable Gate Arrays*, Feb 2017
- S10. **"Vector Acceleration of 1-D DWT Computations using Sparse Matrix Skeletons"** (4 pages)  
 Sidharth Maheshwari, Gourav Modi, Siddhartha, Nachiket Kapre  
*International Conference on Field Programmable Logic and Applications*, Sep 2016
- S9. **"Improving Classification Accuracy of a Machine Learning approach for FPGA Timing Closure"** (4 pages)  
 Que Yanghua, Nachiket Kapre, Harnhua Ng, Kirvy Teo  
*International Symposium on Field Programmable Custom Computing Machines*, May 2016
- S8. **"Case for Design-Specific Machine Learning in Timing Closure of FPGA Designs"** (4 pages)  
 Que Yanghua, Chinnakkannu Adaikkal Raj, Harnhua Ng, Kirvy Teo and Nachiket Kapre  
*International Symposium on Field-Programmable Gate Arrays*, Feb 2016
- S7. **"InTime: A Machine Learning Approach for Efficient Selection of FPGA CAD Tool Parameters"** (4 pages)  
 Nachiket Kapre, Harnhua Ng, Kirvy Teo and Jaco Naude  
*International Symposium on Field-Programmable Gate Arrays*, Page 23-26, February 2015
- S6. **"Fanout Decomposition Dataflow Optimizations for FPGA-based Sparse LU Factorization"** (4 pages)  
 Siddhartha, and Nachiket Kapre  
*International Conference on Field-Programmable Technology*, Page 252-255, December 2014
- S5. **"Analysis and Optimization of a Deeply Pipelined FPGA Soft Processor"** (4 pages)  
 Hui Yan Cheah, Suhaib A. Fahmy and Nachiket Kapre  
*International Conference on Field-Programmable Technology*, Page 235-238, December 2014
- S4. **"Heterogeneous Dataflow Architectures for FPGA-based Sparse LU Factorization"** (4 pages)  
 Siddhartha, and Nachiket Kapre  
*International Conference on Field Programmable Logic and Applications*, Page 1-4, September 2014
- S3. **"Breaking Sequential Dependencies in FPGA-based Sparse LU Factorization"** (4 pages)  
 Siddhartha, and Nachiket Kapre  
*IEEE Symposium on Field-Programmable Custom Computing Machines*, Page 60-63, May 2014
- S2. **"MixFX-SCORE: Heterogeneous Fixed-Point Compilation of Dataflow Computations"** (4 pages)  
 Deheng Ye, and Nachiket Kapre  
*IEEE Symposium on Field-Programmable Custom Computing Machines*, Page 206-209, May 2014

- S1. **"Timing Fault Detection in FPGA-based Circuits"** (4 pages)  
Edward Stott, Joshua M. Levine, Peter Y. K. Cheung, and Nachiket Kapre  
*IEEE Symposium on Field-Programmable Custom Computing Machines*, Page 96-99, May 2014

### Workshop Articles

- W5. **"GraphMMU: Memory Management Unit for Sparse Graph Accelerators"** (8 pages)  
Nachiket Kapre, Han Janglei, Andrew Bean, Pradeep Moorthy, and Siddhartha  
*22nd Reconfigurable Architectures Workshop*, co-located with IPDPS, May 2015
- W4. **"Enhancing Speedups for FPGA Accelerated SPICE through Frequency Scaling and Precision Reduction"** (7 pages)  
Lim Hui Hui and Nachiket Kapre  
*22nd Reconfigurable Architectures Workshop*, co-located with IPDPS, May 2015
- W3. **"Limits of Statically Scheduled Token Dataflow Processing"** (8 pages)  
Nachiket Kapre, and Siddhartha  
*4th International Workshop on Data-Flow Execution Models for Extreme Scale Computing* (co-located with PACT 2014), Page 1-8, August 2014
- W2. **"SPICE<sup>2</sup> - A Spatial Parallel Architecture for Accelerating the SPICE Circuit Simulator"** (6 pages)  
Nachiket Kapre, and André DeHon  
*First Workshop on Intersections between Computer Architecture and Reconfigurable Logic*, co-located with MICRO, December 2010
- W1. **"An NoC Traffic Compiler for efficient FPGA Implementation of Parallel Graph Applications"** (8 pages)  
Nachiket Kapre, and André DeHon  
*Reconfigurable Communication-centric Systems-on-Chip* May 2010

### Posters

- P8. **"Enhancing Butterfly Fat Tree NoCs for FPGAs with Lightweight Flow Control"**  
Gurshaant Malik, Nachiket Kapre  
*International Symposium on Field-Programmable Gate Arrays* Feb 2019
- P7. **"FastTrack: Exploiting Fast FPGA Wiring for Implementing NoC Shortcuts"**  
Tushar Krishna, Nachiket Kapre  
*International Symposium on Field-Programmable Gate Arrays* Feb 2018
- P6. **"Evaluating Embedded FPGA Accelerators for Deep Learning Applications"**  
Gopalakrishna Hegde, Siddhartha, Nachiappan Ramasamy, Vamsi Buddha, Nachiket Kapre  
*International Conference on Field-Programmable Custom Computing Machines* May 2016
- P5. **"Communication Optimization for the 16-core Epiphany Floating-Point Processor Array"**  
Siddhartha, Nachiket Kapre  
*International Conference on Field-Programmable Custom Computing Machines* May 2016
- P4. **"Machine-Learning driven Auto-Tuning of High-Level Synthesis for FPGAs"**  
Li Ting, Harri Sapto Wijaya, Nachiket Kapre  
*International Symposium on Field-Programmable Gate Arrays* Feb 2016

P3. **"Sparse Graph Processing with Soft Processors"**

Nachiket Kapre

*International Conference on Field-Programmable Custom Computing Machines* May 2015

P2. **"FPGA Acceleration of Irregular Iterative Computations using Criticality-Aware Dataflow Optimizations"**

Siddhartha, and Nachiket Kapre

*International Symposium on Field-Programmable Gate Arrays* February 2015

P1. **"Measuring Timing Errors in FPGA-based Circuits"**

Joshua M. Levine, Edward Stott, and Nachiket Kapre

*IEEE Workshop on Silicon Errors in Logic - System Effects* April 2014

### Book Chapter

B2. **"Programming FPGA Applications in VHDL"** (129-153)

Nachiket Kapre, and André DeHon

From *Reconfigurable Computing: The Theory and Practice of FPGA-based Computation*

Published by Morgan Kaufmann, Copyright 2008, ISBN-13: 978-0-12-370522-8

By Scott Hauck and André DeHon

B1. **"Accelerating the SPICE Circuit Simulator using an FPGA - A Case Study"** (389-427)

Nachiket Kapre, and André DeHon

From *High Performance Computing using FPGAs*

Published by Springer New York, Copyright 2013, ISBN-13: 978-1-4614-1790-3

By Wim Vanderbauwhere and Khaled Benkrid

### Magazine Articles

M1. Nachiket Kapre, Dirk Walther, and Christof Koch, and André DeHon

**"Saliency on a chip: a digital approach with an FPGA."**

*The Neuromorphic Engineer*, Volume 1, Issue 2, Autumn 2004

### Patents

PAT3. Nachiket Kapre, Kimon Karras, Dmitri Kitariev, Neil Turton, Siddhartha, Stephan Diestelhorst, Thilini Dassanayake **"Coarse grained reconfigurable architecture"**

U.S. Patent No. 12,585,605 B2, assigned to Xilinx, Inc.; filed Dec 2022, granted Mar 2026

PAT2. Benjamin Morcos, Chris Eliasmith, and Nachiket Kapre

**"Digital circuits for evaluating neural engineering framework style neural networks"**

U.S. Patent No. 11,537,856 B2, filed Aug 2019, granted Dec 2022

PAT1. André DeHon and Nachiket Kapre

**"Method and a circuit using an associative calculator for calculating a sequence of non-associative operations"**

U.S. Patent No. 7,991,817 B2, filed Jan 2007, granted Aug 2011

## Selected Talks

- T17. **“Machine Learning for FPGA Design”**  
Invited Talk *ASAP 2025 Special Sessions – Custom Computing in Canada*, Vancouver, July 2025
- T16. **“Machine Learning for FPGA Design”**  
Invited Talk *Jane Street Xcelerate Colloquium*, New York, May 2025
- T15. **“FPGA Architecture + Programming”**  
Invited Tutorial *SPIN Canada Quantum Computing Workshop*, June 2019
- T14. **“Top-Down and Bottom-Up: Recent Trends in FPGA Network-on-Chip Design”**  
Invited Talk *University of Toronto, Tokyo Institute of Technology, National University of Singapore*, Sep/Dec 2018, Jan 2019
- T13. **“Neural Networks on FPGAs and other Embedded Platforms”**  
Tutorial *Tutorial 4: Embedded Systems Week*, Seoul, October 2017
- T12. **“CaffePresso: Accelerating Convolutional Networks on Embedded SoCs”**  
Invited Talk *Carnegie Mellon University, University of Toronto, Seoul National University*, Oct/Nov 2016, Oct 2018.
- T11. **“OpenCL for CPUs, GPUs, and FPGAs”**  
Invited Lecturer *Harbin Summer School 2017*, Harbin University of Technology, Harbin, China, July 2017
- T10. **“OpenCL for FPGAs and Neural Networks”**  
Invited Lecturer *Nengo Summer School 2017*, University of Waterloo, June 2017
- T9. **“OpenCL for CPUs, GPUs, and FPGAs”**  
Invited Lecturer *PAPAA Summer School 2016*, University of Hong Kong, July 2016
- T8. **“A Case for Embedded FPGA-based SoCs for Energy-Efficient Acceleration of Graph Problems”**  
*Supercomputing Frontiers 2015*, A\*Star, Singapore, March 2015
- T7. **“Assessment Engineering for High-Impact, Programming-Centric Courses in Science and Technology”** and **“Continuous Automated Assessment for Project-Centric Learning”** at the *Good to Great* teaching workshops in 2014 and 2015, NTU Singapore.
- T6. **“FX-SCORE Precision Analysis”, and “Libraries for NVIDIA GPUs”**  
Invited Lecturer *PAPAA Summer School 2014*, University of Hong Kong, July 2014
- T5. **“Rebooting SCORE for the Next Generation”**  
Invited Talk at *ASPLOS 2012* as part of *CCPC Workshop*, London, March 2012, and Guest Lecture for *Custom Computing CO108* (Imperial College), March 2012
- T4. **“Parallelizing SPICE using FPGAs”** Invited speaker at the *ASC 2012 Summer School*, Imperial College London.

- T3. **“Accelerating SPICE using FPGAs: A Retrospective and Vision for the Future”**  
Research Seminars at *University of Southampton, Oxford, University of Glasgow, University of York, National University of Singapore, Mahanakorn University of Technology* (2011-2013).
- T2. **“Spatial SPICE Mapping and Lessons”** (work-in-progress)  
CASCADIA Workshop, *University of British Columbia (UBC)*, Vancouver Canada, August 2010,  
IBM, Inc., Austin USA, August 2009,  
*Indian Institute of Science*, Bengaluru India, March 2010,  
*Xilinx, Inc*, San Jose USA, February 2009.
- T1. **“Exploiting Application Structure in On-Chip Network Design”**  
Gent, Belgium *University of Gent* and Munich, Germany *TU Munich* July-August 2007.

## Open Source

I encourage my students to actively publish their code and design artifacts online for community sharing. Open source code allows other researchers to replicate our results, improve the quality of the artifact, and advance the community forward. Since 2013, I have open sourced 10 code repositories to accompany some of my papers.

- R13. FPGA Device Driver, <https://github.com/vipinkmenon/fpgadriver>, FPT 2013, (2 forks, 2 stars)
- R12. BitGPU CUDA Code, <https://github.com/yedeheng/bitgpu>, FPGA 2015, (3 forks, 8 stars)
- R11. Graph Access Memory Controller, [https://bitbucket.org/nachiketkapre/graph\\_mmu](https://bitbucket.org/nachiketkapre/graph_mmu), RAW 2015
- R10. CaffePresso, <https://github.com/gplhegde/caffepresso>, CASES 2016, (23 forks, 73 stars, 15 watches)
- R9. Plunify ML Algorithms, <https://bitbucket.org/nachiketkapre/plunify-ml>, FPL 2016
- R8. Butterfly Fat Tree RTL, <https://bitbucket.org/nachiketkapre/bft-deflection>, FPL 2017
- R7. HopliteRT Bounds calculation tool, <https://git.uwaterloo.ca/watcag-public/hopliter-bounds>, FPT 2017
- R6. HopliteBuf Bounds calculation tool, <https://git.uwaterloo.ca/watcag-public/hoplitebuf-bounds>, FPGA 2019 (Best Paper Nominee)
- R5. BFT Flow RTL, <https://git.uwaterloo.ca/watcag-public/bft-flow>, FCCM 2019
- R4. RapidRoute CAD tool, <https://git.uwaterloo.ca/watcag-public/RapidRoute>, FCCM 2019 (Best Short Paper Runner Up) + FPL 2019
- R3. Scaling-the-Cascades RTL, <https://git.uwaterloo.ca/watcag-public/fpga-cascades-rtl>, FPL 2019.
- R2. Systolic Array Partitioning Tool, <https://git.uwaterloo.ca/watcag-public/fpga-syspart>, FPT 2019.
- R1. RapidLayout Floorplanning Tool, <https://git.uwaterloo.ca/watcag-public/rapidlayout>, FPL 2020

## Professional Activities

There are four prestigious conferences in the FPGA/reconfigurable computing field: FPGA, FCCM, FPL, and FPT. The membership on the Program Committees is based by reputation and academic track record. For instance, the FPGA TPC (Technical Program Committee) is the most difficult to get invited to, and is a mark of community recognition for your work by top academics and industry researchers. Typically, each PC member will review  $\approx 10$  papers and be expected to discuss these papers at a PC meeting.

An invitation to serve as Program Chair/Co-Chair to any of the top four conferences is an opportunity to lead. Each of these conferences typically attract 150–250 attendees each year and attended by top academics, graduate students, industrial researchers and practitioners.

I was selected to chair the TCFPGA **Hall of Fame** Selection Committee for the Class of 2019 through a voting process. The ACM SIGDA Technical Committee on FPGAs and Reconfigurable Computing (TCFPGA) is responsible for “promoting FPGA and Reconfigurable Computing Community”. I have been a member of the IEEE and ACM since 2003.

### Conference Chairing Duties:

1. Technical Program Chair for IEEE International Symposium On Field-Programmable Custom Computing Machines 2023.
2. Technical Program Co-Chair for IEEE International Conference on Field-Programmable Technology 2015.
3. Technical Program Co-Chair for International Symposium on Highly-Efficient Accelerators and Reconfigurable Technologies, 2018

### Conference Program Committee Memberships at top-tier conferences:

1. Program Committee Member for ACM/SIGDA International Symposium on Field-Programmable Gate Arrays (most prestigious conference) 2017, 2018, 2019, 2020, 2021, 2022, 2023.
2. Program Committee Member for IEEE International Conference on Field-Programmable Custom Computing Machines 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023.
3. Program Committee Member for IEEE International Conference on Field-Programmable Technology 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022.
4. Program Committee Member for International Conference on Field-Programmable Logic and Application 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022.
5. Program Committee Member for Design, Automation, and Test in Europe for the Reconfigurable Computing track 2018, 2019.

### Other Program Committee Memberships:

1. Program Committee Member for ASAP 2015, 2016, 2017, 2018.
2. Program Committee Member for RAW workshop 2015, 2016.
3. Program Committee Member for RECOSOC 2012, 2013.
4. Program Committee Member for HEART 2012, 2013, 2014, 2015, 2016, 2017, 2018.

Journal Reviewing Responsibilities:

1. Reviewer ACM Transactions on Reconfigurable Technology and Systems, 36 papers
2. Reviewer ACM Transactions on Computer-Aided Design of Integrated Circuits and Systems, 12 reviews
3. Reviewer IEEE Transactions on Biomedical Circuits and Systems, 1 review

Consulting Roles:

Maxeler Inc. July 2011–July 2012, Commercializing compilation aspects of PhD work.

Plunify Inc. 2014-present, Consulting on formulation of ML algorithm for driving FPGA CAD tools.

## Funding and Grants

| Grant                                                                 | Role              | Title                                                                                                                                                                                                | Funding                       | Value          | Time      |
|-----------------------------------------------------------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|----------------|-----------|
| <b>NSERC Discovery Grant Renewal</b>                                  | PI                | Rethinking Configurable Logic Architectures in the age of AI Proposal rated as <b>Very Strong</b> for Excellence of Researcher and Merit of Proposal.                                                | CAD\$265K (100%)              | 1 PhD          | 2024-2029 |
| <b>MITACS Globalink</b>                                               | PI                | High-Level Synthesis of NENGO Neural Networks                                                                                                                                                        | CAD\$15K (7.5K from ABC Inc.) | 1 intern       | 2020-2021 |
| <b>Huawei</b> (U of T lead institution – Jason Anderson, Vaughn Betz) | PI (Sub-contract) | CGRA and Embedded FPGA architecture and tools                                                                                                                                                        | CAD\$157.5K (100%)            | 1 PhD + 1 MASc | 2019-2021 |
| <b>RTI</b> (Led by Prof Slim Boumaiza)                                | Co-PI             | Real-Time Baseband Signal Processing and Reconfigurable Computing Development Platform                                                                                                               | CAD\$148,905                  | FPGA system    | 2019-2020 |
| <b>Xilinx</b>                                                         | PI                | Unrestricted Gift to support research activities in group                                                                                                                                            | CAD\$10K (100%)               | 1 PhD term     | 2018-2019 |
| <b>NSERC Discovery Grant</b>                                          | PI                | Enhancing Hardware Compilation for Reconfigurable Architectures using Machine Learning and Cloud Computing. Proposal rated as <b>Very Strong</b> for Excellence of Researcher and Merit of Proposal. | CAD\$165K (100%)              | 1 PhD          | 2018-2022 |
| <b>Waterloo Startup</b>                                               | PI                | Startup Activities                                                                                                                                                                                   | CAD\$150K (100%)              | -              | 2016–now  |

| Grant                      | Role    | Title                                                                                                           | Funding        | Value     | Time      |
|----------------------------|---------|-----------------------------------------------------------------------------------------------------------------|----------------|-----------|-----------|
| AcRF Tier1 Grant           | PI      | Intelligent CAD Algorithms using Cloud Computing and Machine Learning                                           | S\$100K (100%) | 1 RA      | 2015-2016 |
| MIT-SMART Innovation Grant | Co-PI   | Ultra Low-Power Embedded Machine Learner for Image/Video Processing in Robotics and Portable / Wearable Devices | S\$249K (20%)  | 0.5 RA    | 2015-2017 |
| Delta Electronics Grant    | Co-PI   | Low Power Embedded Platform for Machine Learning in IoT                                                         | S\$586K (17%)  | 0.5 RA    | 2015-2017 |
| CELT Edex                  | PI      | Cloud-enhanced Learning for Hardware-Oriented Labs in Computing                                                 | S\$37K (100%)  | 1 UG      | 2015      |
| CELT Edex                  | PI      | Continuous Automated Assessment for Project-Centric Learning                                                    | S\$40K (100%)  | 1 UG      | 2014      |
| Relax FPGA Equipment       | Collab. | Earth Observatory of Singapore Grant                                                                            | N/A            | FPGA card | 2013-2014 |
| CELT Edex                  | PI      | Assessment Engineering for High-Impact, Programming-Centric Courses in Science and Technology                   | S\$25K (100%)  | 1 UG      | 2013      |
| AcRF Tier1 Grant           | PI      | Parallel Sparse-Graph Accelerators for Electronic Design Automation Algorithms                                  | S\$150K (100%) | 1 PhD     | 2013-2014 |
| NTU COE Seed Grant         | PI      | Outreach activities to launch larger grants                                                                     | S\$50K (100%)  | N/A       | 2012      |
| NTU Startup Grant          | PI      | Scalability and Energy-Efficiency in Next-Generation Spatial Computer Architectures                             | S\$100K        | 1 PhD     | 2012-2015 |
| NSERC Strategic Grant      | Collab. | Compute-Oriented FPGA Device Architectures and Tools                                                            | CAD\$290K (0%) | 2010-2013 |           |

## Teaching

Summary of courses taught.

| Class #    | Course Title                                                  | Level     | # of Students | Univ.    |
|------------|---------------------------------------------------------------|-----------|---------------|----------|
| ECE 327    | Digital Systems Engineering                                   | Undergrad | 130-170       | Waterloo |
| ECE 627    | Register-transfer-level Digital Systems                       | Grad      | ≈15-20        | Waterloo |
| ECE 720 T6 | Trends in Reconfigurable Computing                            | Grad      | ≈10           | Waterloo |
| CE 4052    | Embedded Software Development                                 | Undergrad | ≈10-20        | NTU      |
| CE 4054    | Programmable System-on-Chip                                   | Undergrad | ≈5-20         | NTU      |
| ES 6152    | Embedded Software Development                                 | Grad      | ≈15-40        | NTU      |
| ES 6154    | Programmable System-on-Chip                                   | Grad      | ≈35-50        | NTU      |
| ES 7501    | Electronic Design Automation                                  | Grad      | 5             | NTU      |
| CE 7451    | Research Methods in Computer Science and Computer Engineering | Grad      | 20            | NTU      |

Course evaluation for undergraduate and graduate classes taught in Waterloo. The older course evaluation framework before 2021 is in the following format. The older course evaluation framework before 2021 is in the following format.

| Term & Year | Class # | Course Title                | # of Stud. | Q1-3 | Q4-6 | Resp % | Development                                     |
|-------------|---------|-----------------------------|------------|------|------|--------|-------------------------------------------------|
| Winter 2025 | ECE 327 | Digital Systems Engineering | 168        | 4.22 | 3.98 | 60.7%  | LLM-on-an-FPGA labs and Crowdmark-online exams. |
| Winter 2024 | ECE 327 | Digital Systems Engineering | 141        | 4.53 | 4.28 | 73.1%  | First cut of LLM-on-an-FPGA labs.               |
| Spring 2023 | ECE 327 | Digital Systems Engineering | 157        | 4.60 | 4.52 | 29.3%  | None.                                           |

Courses evaluation for classes taught at NTU, Singapore. Evaluation scores here are a single number out of 5. Two sets of scores indicate undergraduate and graduate student ratings for the cross-listed courses.

At NTU, I developed two final year undergraduate courses that were cross-listed with graduate program. They were CE4054 (ES6154), and CE4052 (ES6152) as listed in the tables above. These courses involved a significant hands-on lab/project component. I developed the lectures, as well as labs for these courses. CE4052 (ES6152) involved developing OpenCV applications on Android tablets and optimizing their implementation. CE4054 (ES6154) involved developing hardware-software codesign projects involving the Xilinx Zynq SoC. I am particularly proud of the highly-reviewed CE7451 Research Methods course that I co-taught once with three other faculty. It introduced PhD students to the research process and tools and techniques that could help them along their journey.

At Waterloo, I developed a graduate course in Reconfigurable Computing ECE720T6 that introduced FPGA architecture, FPGA CAD tools, and the OpenCL programming language to the students. I also took Mark Aagaard's ECE327 course outline and revamped the lectures, switched programming language to Verilog, redeveloped labs for the course, switched to an upgraded Xilinx Pynq FPGA board, and moved assessment to a two-quiz system instead of a midterm. In 2018–2021, every year, I received a letter from the Dean of Engineering for the high review scores this course received from the students.

| Term & Year | Class # | Course Title                | # of Stud. | Q1-9    | Q10       | Resp %   | Development                                       |
|-------------|---------|-----------------------------|------------|---------|-----------|----------|---------------------------------------------------|
| Spring 2021 | ECE 327 | Digital Systems Engineering | 181        | 88-99   | 97        | 74%      | Online/Remote learning videos + Lab improvements. |
| Spring 2020 | ECE 327 | Digital Systems Engineering | 189        | 81-99   | 96        | 66-67%   | Online/Remote learning videos + Lab improvements. |
| Spring 2019 | ECE 327 | Digital Systems Engineering | 169        | 82-98   | 92-96     | 67-69%   | Verilog + New labs + New FPGA.                    |
| Spring 2018 | ECE 327 | Digital Systems Engineering | 162        | 93.5-95 | 92.9-96.2 | 50-55.2% | New tutorials.                                    |
| Spring 2017 | ECE 327 | Digital Systems Engineering | 132        | 81      | 72.5      | 53%      | Revamp lectures.                                  |

| Term & Year | Class #   | Course Title                            | # of Stud. | Q1-3 | Q4-6 | Resp % | Development                                   |
|-------------|-----------|-----------------------------------------|------------|------|------|--------|-----------------------------------------------|
| Fall 2024   | ECE 720T7 | On-Chip Interconnect                    | 20         | 4.61 | 4.15 | 55%    | Brand-new course and brand-new hands-on labs. |
| Fall 2023   | ECE 722   | FPGA Architecture                       | 12         | 4.67 | 3.82 | 91%    | Minor lab cleanup.                            |
| Fall 2023   | ECE 627   | Register-Transfer Level Digital Systems | 26         | 4.73 | 4.73 | 73%    | None.                                         |

## Advising and Supervision

### Current PhD Students:

Emir Guevara (Waterloo), Fall 2025–present.

Ahmed Elsousy (Waterloo), Fall 2025–present, co-supervised with Prof. Andrew Boutros.

Thomas Badros (Waterloo), Fall 2025–present, co-supervised with Prof. Andrew Boutros.

### Current MASc Students:

Jehanzeb Mirza (Waterloo), Winter 2023–now, (co-supervised with Prof. Vincent Gaudet), recipient of the Queen Elizabeth II Graduate Scholarship in Science and Technology (QEII-GSST).

### Current URA/USRA Students:

| Term & Year | Class #    | Course Title                            | # of Stud. | Q1-8 | Q9   | Resp. % | Development                        |
|-------------|------------|-----------------------------------------|------------|------|------|---------|------------------------------------|
| Winter 2021 | ECE 722    | Reconfigurable Computing                | 14         | 86.8 | 83.4 | 64.3%   | New quizzes, labs, guest lectures. |
| Winter 2019 | ECE 627    | Register-transfer-level Digital Systems | 19         | 93.8 | 93.1 | 94.7%   | New labs.                          |
| Winter 2018 | ECE 627    | Register-transfer-level Digital Systems | 17         | 74.8 | 68.8 | 94.1%   | Verilog + New labs + FPGA.         |
| Winter 2018 | ECE 720 T6 | Trends in Reconfigurable Computing      | 9          | 65.5 | 66.7 | 100%    | New projects.                      |
| Winter 2017 | ECE 720 T6 | Trends in Reconfigurable Computing      | 7          | 90.2 | 100  | 57.1%   | New course.                        |

Elias Kountouris

Jonatham Lam

Lakshya Rao

Cameron Wang

#### Past PhD Students:

Gurshaant Malik (Waterloo), Fall 2017-Spring 2025, solely supervised, PhD 2025.

*Current Position:* Engineer, Meta.

**“Learning Design Parameters to Build Application Customizable Network-on-Chips for FPGAs”,**

<https://hdl.handle.net/10012/21388>

Siddhartha (NTU, Jan 2013-Jan 2019, solely supervised), PhD 2019.

*Current Position:* AMD Research Singapore.

*Previous Position:* Postdoc at University of Sydney.

**“Dataflow optimized overlays for FPGAs”,**

<https://dr.ntu.edu.sg/handle/10220/47803>

Cheah Hui Yan (NTU, 2014–2016, co-supervised with Suhaib Fahmy), PhD 2016.

*Current Position:* Engineer at Xilinx, Singapore.

**“The iDEA architecture-focused FPGA soft processor”,**

<https://repository.ntu.edu.sg/handle/10356/68990>

Abid Rafique (2010-2013, co-supervised with George Constantinides, Imperial College), PhD 2013.

**“Communication Optimization in Iterative Numerical Algorithms: An Algorithm-Architecture Interaction”,**

<https://spiral.imperial.ac.uk/handle/10044/1/17837>

| Term & Year | Class #        | Course Title                                                  | # of Stud. | Eval.      | Resp. % |
|-------------|----------------|---------------------------------------------------------------|------------|------------|---------|
| Winter 2013 | CE4054, ES6154 | Programmable System-on-Chip                                   | 28         | 4.36, 4.44 | 78%     |
| Winter 2014 | CE4054, ES6154 | Programmable System-on-Chip                                   | 70         | 4.32, 4.49 | 58%     |
| Winter 2015 | CE4054, ES6154 | Programmable System-on-Chip                                   | 48         | 4.21, 4.59 | 27%     |
| Fall 2013   | CE4052, ES6152 | Embedded Software Development                                 | 43         | 3.4, 3.73  | 64%     |
| Fall 2014   | CE4052, ES6152 | Embedded Software Development                                 | 66         | 4.39, 4.44 | 72%     |
| Fall 2015   | CE4052, ES6152 | Embedded Software Development                                 | 52         | 4.51, 4.36 | 67%     |
| Fall 2013   | CE7451         | Research Methods in Computer Science and Computer Engineering | 19         | 4.53       | 73%     |
| Fall 2013   | ES7501         | Electronic Design Automation                                  | 5          | 4.36       | 80%     |

Ye Deheng (NTU, 2012-2016, co-supervised with Zhenchang Xing), PhD 2016.  
*Current Position:* Engineer at Tencent, China.

Andrew Bean (Imperial College), PhD 2016. (collaborator)

#### Past MASc Students:

Marat Bekmyrza (Waterloo) Fall 2021-Spring 2025. (co-supervised with Prof. Hiren Patel).  
*Current Position:* Engineer at Synopsys, Toronto.  
**"Language Model Inference on FPGA with Integer-only Operations"**  
<https://hdl.handle.net/10012/21600>

Aravind Vellora Valayapara (Waterloo) Fall 2021-Fall 2023 (co-supervised with Prof Seyed Zahedi).  
*Current Position:* Compiler Engineer, Groq.  
**"Fairness Notions on Hardware Resource Configuration"**  
<http://hdl.handle.net/10012/20063>

Omidi, MohammadHadi (Waterloo) Fall 2021-Winter 2024 (co-supervised with Prof Seyed Zahedi).  
*Current Position:* Compiler Engineer, Groq.  
**"Fair and Efficient Resource Scheduling in Heterogeneous Multi-Agent Systems"**  
<http://hdl.handle.net/10012/20362>

Long Chan Chan (Waterloo) (Spring 2020-Winter 2021),  
*Current Position:* Engineer at Efinix, Toronto.  
**"Implementing FPGA-optimized Systolic Arrays using 2D Knapsack and Evolutionary Algorithms",**

<https://uwspace.uwaterloo.ca/handle/10012/17969>

Alireza Mellat (Waterloo) (Winter 2020-Winter 2021),  
*Current Position:* Engineer at Microchip, Toronto.

**“Mocarabe: High-Performance Time-Multiplexed Overlays for FPGAs”**,  
<https://uwspace.uwaterloo.ca/handle/10012/17993>

Srinirdheeshwar Kuttuva Prakash (Waterloo) (Fall 2019-Winter 2021),  
*Current Position:* Engineer at Microchip, Toronto.

**“Managing HBM’s bandwidth in Multi-Die FPGAs using Overlay NoCs”**,  
<https://uwspace.uwaterloo.ca/handle/10012/17909>

Ian Elmor Lang (Waterloo) (Fall 2019-Winter 2021),  
*Current Position:* Engineer at Intel FPGA, Toronto.

**“Worst-Case Latency Analysis for the Versal Network-on-Chip”**,  
<https://uwspace.uwaterloo.ca/handle/10012/17910>

Tushar Garg (Waterloo) (Fall 2017-Winter 2019),  
*Current Position:* Engineer at Microsoft Azure FPGA Team, Seattle.

**“HopliteBuf FPGA Network-on-Chip: Architecture and Analysis”**,  
<https://uwspace.uwaterloo.ca/handle/10012/14543>

Benjamin Morcos (Waterloo) (Fall 2017-Spring 2019), part-time.  
*Current Position:* Engineer at ABR, Waterloo.

**“NengoFPGA: an FPGA Backend for the Nengo Neural Simulator”**,  
<https://uwspace.uwaterloo.ca/handle/10012/14923>

Chethan Kumar Basavaraju, Nachiappan Ramasamy, Gourav Modi, Joseph George, Jacob Ginu, Manoj Venkat (NTU) MSc 2015-16. They work in hardware and embedded design companies in Singapore.

Adaikkal Raj, Selva Kumar Jayakrishnan, Jianrong, Swetha Venugopal, Kiran Ganapathi, Kunal Gokhale, (NTU) MSc 2014-15

Kanchan Kaur, (NTU) MSc 2013-14

Helene Martorell, Emmanouil Spanakis, Fang Zhou, Wei Lizhong (Imperial College), MSc 2011.

#### Past UG Students:

Gavin Lusby (Winter 2025 URA).

Raymond Wu (Winter 2025 URA).

Elias Kountouris (Fall 2023 URA, Spring 2025 URA).

Lakshya Rao (Fall 2023 URA, Spring 2025 URA).

John Jekel (Winter 2023 URA, Fall 2024 URA).

Evan Lin (Spring 2025 USRA).

Kevin Song (Spring 2020 URA, Fall 2020 USRA).

Justin Borromeo (Spring 2020).

Amanda Redpath, Caroline White (Winter 2019)

Fred Tombs, Harry Chan (Spring 2019, Winter 2020).

Jay Weng, Leo Liu (Spring 2018).

Almeet Bhullar, Alex Labach, Asif Islam (Spring 2017).

Shubham Agarwal (NTU), Bachelor's 2016.  
**"Networks on chip for FPGAs"**,  
<https://repository.ntu.edu.sg/handle/10356/66694>,

Que Yanghua (NTU), Bachelor's 2015,  
**"Machine learning on FPGA CAD"**, <https://repository.ntu.edu.sg/handle/10356/65738>.  
Currently at Schlumberger, Singapore

Pradeep Moorthy (NTU), Bachelor's 2014.  
*Current Position:* Engineer at NVIDIA, MSc U of Toronto,  
**"Network design and characterization for ZedWulf"**,  
<https://repository.ntu.edu.sg/handle/10356/61959>,

Han Jianglei (NTU), Bachelor's 2014.  
*Current Position:* PhD Student at MIT SMART, Singapore,  
**"Graph MMU design for Zedwulf"**,  
<https://repository.ntu.edu.sg/handle/10356/61961>,

Soh Jun Jie (NTU), Bachelor's 2014.  
**"Soft core and hard core vector processors using vector backend"**,  
<https://repository.ntu.edu.sg/handle/10356/59019>,

Lim Hui Hui (NTU), Bachelor's 2013.  
**"Exploring algorithmic error tolerance of circuit simulators"**,  
<https://repository.ntu.edu.sg/handle/10356/55040>,

Siddhartha and Ramitha (Imperial College), BEng 2012, Dulitha Gunasekara (Imperial College), MEng 2012.

Coryan Wilson-Shah (Cambridge/Imperial College), UROP 2011, Cody Huang (UC Davis/Imperial College), Exchange Student 2011.

## Service

### Committee Memberships:

1. Departmental Faculty Hiring Committee: Hardware DACA 2018-2019 cycle.
2. PhD Examination Committee: Background Examiner for Sheida Gohardehi (July 2019), Reza Mirosanlou (2018)
3. PhD Examination Committee: Examiner for Comprehensive for Jonathan Mohamed Shahen (2018), Anirudh Kaushik (2017), Cui Ying (NTU, 2012), Yang Liwei (NTU, 2014).
4. PhD Examination Committee: Chair for Benyamin Ghogh (2018), Silu Chen (Chemistry, 2018), Mostafa Alizadeh (2017), Golnaz Zhanjani (2016).
5. External PhD Examiner: Alex Rodionov (University of Toronto, June 2019), Haomiao Wang (University of Auckland, Feb 2019)

### Direction of Labs:

1. Waterloo Configurable Architectures Group (WatCAG), Lab Head (2016-present)

### Membership of Centers:

1. Center for Theoretical Neuroscience (CTN), University of Waterloo (2018-present)

## Awards and Honors

**Best Paper Award**, ACM Transactions on Reconfigurable Technology and Systems 2023

**Best Paper Award**, International Conference on Field-Programmable Technology 2019.

**Best Paper Award**, ACM Transactions on Reconfigurable Technology and Systems 2017

**Best Paper Award**, International Conference on Compilers, Architecture, and Synthesis for Embedded Systems 2016

**Best Paper Award**, International Conference on Field-Programmable Logic and Applications 2015

**HiPEAC Paper Award, 2013**, For the FCCM 2013 paper ‘Application Composition and Communication Optimization of Iterative Solvers using FPGAs’

**FCCM20 25-most Influential Papers Award at FCCM 2013**, IEEE Symposium on Field-Programmable Custom Computing Machines 2006

**Best Paper Award**, International Conference on Field-Programmable Technology 2011

**Imperial College Junior Research Fellowship (2010-2012)** awarded through a competitive recruitment process